

Japman Singh

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EXPERIENCE

Summer Research Intern, Annam.ai

June – July 2026

Wind-Sensing Research Project

IIT Ropar

- Contributed to a wind-sensing research project, writing and debugging embedded C firmware for ultrasonic transducers on TI MSP430 microcontrollers to measure Time-of-Flight (ToF) and estimate wind speed and direction
- Implemented signal averaging and filtering to improve sensor measurement stability, and transmitted processed data via UART for real-time analysis

Software Intern, iHub – AWaDH @ IIT Ropar

June – July 2025

Project: Soil Spectra

IIT Ropar

- Preprocessed a 19,000+ sample hyperspectral soil reflectance dataset (1,000 spectral features, 400–940 nm) using Python, Pandas, and NumPy, performing duplicate removal, outlier treatment, missing value handling, and spectral scaling
- Optimized interpolation step size and data cleaning pipelines for soil nutrient prediction models, achieving R^2 of 0.803 (Organic Carbon) and 0.690 (Nitrogen) using scikit-learn
- Evaluated multiple regression models to predict soil nutrients (Organic Carbon, Nitrogen, Phosphorus, Potassium) from spectral reflectance data, analyzing RMSE and R^2 across nutrients to identify best-fit configurations
- Collaborated within a cross-functional team (Android + AI/ML) building an end-to-end IoT-based precision agriculture system with BLE-connected optical soil sensors

TECHNICAL SKILLS

Languages: Java, JavaScript, HTML, CSS

Frameworks & Libraries: React, Node.js, Express.js, Tailwind CSS, Next.js

Database & Networking: MongoDB, Redis, TCP/IP, HTTPS

Developer Tools: Git, VS Code, IntelliJ IDEA, Postman, MongoDB Compass

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming, RESTful APIs, MVC Architecture

Competitive Programming: 400+ DSA problems solved across LeetCode ([im_japmansingh](https://leetcode.com/u/im_japmansingh/), rating 1652), Codeforces ([japman006](https://codeforces.com/profile/japman006)), and GeeksforGeeks

PROJECTS

DevBoard

Full-Stack Developer Productivity Platform

React, Node.js, Express, MongoDB

- Architected a full-stack MERN platform with a modular MVC backend, exposing 15+ RESTful endpoints to unify task management, notes, and third-party coding-platform data into a single dashboard
- Built data-normalization services aggregating LeetCode, Codeforces, and GitHub APIs into a unified schema, reducing manual profile-checking across 3 platforms into a single real-time dashboard view
- Implemented JWT-based authentication with bcrypt hashing and route-level middleware, securing all user sessions and API endpoints against unauthorized access

API Rate Limiter

In Progress

Distributed Request-Throttling Middleware

Node.js, Express, Redis, MongoDB

- Building a distributed API rate limiter using the token bucket algorithm, enforced as reusable Express middleware to throttle per-client request rates
- Using Redis for atomic bucket-state tracking, ensuring accurate limits under concurrent requests across multiple server instances without race conditions

EDUCATION

Punjabi University

B.Tech in Computer Science

Patiala, Punjab, India

Aug. 2023 – Aug. 2027